

Detailed Reviewer-by-Reviewer Response and Old-vs-New Revision Audit

Manuscript: *Energy Infrastructure for AI Data Centers: A Critical Review of Grid Integration, Techno-Economic Challenges, and Future Pathways*

Reviewer evaluated: Reviewer #2

Documents compared: the original manuscript (... - Old.docx) and the current uploaded revised manuscript (... (1) .docx).

Special emphasis: Word-highlighted revisions in the current manuscript.

Important conclusion before using this as a response letter

The current revised manuscript is **substantially stronger than the original**, especially in the highlighted revisions to the review methodology, storage-sizing discussion, seven-component framework, workload flexibility, economic analysis, and limitations. However, the current uploaded file does **not** yet support several claims made in the older Reviewer_Response_Letter.docx.

The current upload contains **747 body paragraphs, 30 tables, and 334 highlighted body paragraphs**. The largest highlighted revision clusters are in Section 5.3 (storage sizing), Sections 7.3–7.5 (technology selection, electrical architecture, and control), Section 7.8 (economic model), Sections 10.1–10.5 (economics), Section 1.6 (review methodology), and Section 13 (limitations).

Before resubmission, the response letter and manuscript should be

synchronized. In particular, the current file still has three reviewer-identified typos, lacks the required declarations/highlights/nomenclature, lacks explicit source lines in most figure/table captions, does not report the exact search strings or last search date, and still contains a conceptual availability/LOLE conflation. The current bibliography also appears to contain **178 numbered entries**, while the manuscript states that **185 sources** were included in the qualitative synthesis; this must be reconciled before claiming the count issue is fully resolved.

Status matrix

Reviewer comment	Current status
2.1	PARTIALLY ADDRESSED
2.2	PARTIALLY ADDRESSED
2.3	ADDRESSED
2.4	ADDRESSED
2.5	ADDRESSED
2.6	ADDRESSED
2.7	ADDRESSED
2.8	PARTIALLY ADDRESSED — IMPORTANT
2.9	ADDRESSED
2.10	ADDRESSED
2.11	ADDRESSED WITH CLARIFICATION
2.12	SUBSTANTIALLY ADDRESSED
2.13	ADDRESSED
2.14	ADDRESSED FOR THE THREE NAMED CITATIONS
2.15	SUBSTANTIALLY ADDRESSED
2.16	ADDRESSED

2.17	ADDRESSED
2.18	PARTIALLY ADDRESSED
2.19	ADDRESSED
2.20	ADDRESSED
2.21	ADDRESSED
2.22	SUBSTANTIALLY ADDRESSED — LANGUAGE CLEANUP NEEDED
2.23	NOT YET FULLY ADDRESSED
2.24	PARTIALLY ADDRESSED
2.25	ADDRESSED
2.26	NOT ADDRESSED IN THE CURRENT UPLOAD
2.27	PARTIALLY ADDRESSED
2.28	NOT ADDRESSED IN THE CURRENT UPLOAD
2.29	SUBSTANTIALLY ADDRESSED

Recommended General Response to Reviewer #2

We thank Reviewer #2 for the detailed and technically rigorous evaluation of the manuscript. The comments led to substantial revision of both the content and the presentation of the review. In particular, we added a structured review-methodology section and PRISMA-style screening figure; reconciled the evidence-base description; removed unsupported original numerical outputs; corrected the deterministic and LOLE formulations; distinguished battery-pack prices from installed BESS costs; separated generation-only LCOE, storage-service cost, partially firm hybrid cost, and delivered-grid electricity; expanded the seven-component framework

with explicit inputs, decision rules, outputs, and modeling requirements; added a dedicated treatment of workload-flexibility constraints and the water–energy tradeoff; expanded the limitations and critical-appraisal discussion; and revised the economic analysis so that published benchmarks are not presented as universal project results.

The revised manuscript also makes clearer that the proposed seven-component framework is a screening and decision-support structure rather than a site-specific design optimization. Throughout the revision, we have sought to distinguish literature-derived benchmarks from project-specific estimates and to identify where chronological, site-specific analysis is required. The detailed responses below explain each change relative to the original manuscript and identify the corresponding location in the revised version.

Do not send the paragraph above as a statement that every point is already closed. Complete the items marked **PARTIALLY ADDRESSED** or **NOT YET FULLY ADDRESSED** below first.

Comment 2.1

Reviewer comment

No review methodology is reported. Please add a PRISMA-style methodology with the databases searched, the search strings used, the date window, the inclusion and exclusion criteria, the screening counts (n identified, n screened, n excluded with reasons, n included), and a PRISMA flow diagram.

Original manuscript

The original manuscript contained no dedicated review-methodology section. It did not document a reproducible search strategy, eligibility criteria, screening stages, or a PRISMA flow diagram.

Current revised manuscript

The current manuscript adds highlighted Sections **1.6, 1.6.1, 1.6.2, and 1.6.3**. Section 1.6.1 identifies Scopus, Web of Science Core Collection, and IEEE Xplore as the main bibliographic databases, with ScienceDirect used for full-text discovery and Google Scholar as a supplementary discovery/citation-searching resource. It states a primary review window of 2015–2026 and explains why earlier foundational sources may be retained. Section 1.6.2 gives inclusion and exclusion logic, and Section 1.6.3 states the final qualitative-synthesis count and source-type breakdown.

The new Figure 2 visibly reports: 1,247 database records + 184 additional records = 1,431 identified; 1,108 after duplicate removal; 1,108 screened; 723 excluded at title/abstract screening; 385 full texts assessed; 200 full texts excluded; and 185 studies included. The figure also gives full-text exclusion categories.

Highlighted revision evidence

The methodology additions are strongly highlighted in the Word file, particularly Sections 1.6.1–1.6.3. Figure 2 is a newly added PRISMA-style infographic.

Assessment

PARTIALLY ADDRESSED

This is a major improvement, but the reviewer explicitly requested the **search strings** and **date window/search date**. The current prose gives search concepts and the publication window, but it does not report the exact database-specific Boolean strings or the actual last search date. Figure 2 states that Boolean strings were used, but that is not the same as reporting them. The wording “follows the principles of PRISMA 2020” is also safer than claiming complete PRISMA compliance unless all required items are documented.

Required final action before resubmission

Add the real, preserved search strings for each database (or a transparent database-specific adaptation table) and the actual last search date. Do **not** invent either. If exact strings were not preserved, describe the search concept groups and explicitly state that database syntax was adapted; do not fabricate a verbatim search history.

Recommended response to the reviewer

We agree that an explicit review methodology is necessary. Section 1.6 has been added and is organized into Evidence Selection, Eligibility Criteria, and Screening and Selection Process. The revised section identifies the principal bibliographic databases and supplementary discovery sources, defines the 2015–2026 primary publication window, states the inclusion and exclusion criteria, and reports the screening process. Figure 2 now presents the review flow from 1,431 identified records to 185 sources included in the final qualitative synthesis. We have also reported the database-specific search strategy and the actual last search date as **[INSERT VERIFIED SEARCH DETAILS / DATE]**. Because the review is a mixed-evidence critical review rather than a quantitative systematic meta-analysis, we describe the process as PRISMA 2020-informed and use PRISMA to improve transparency of identification, screening, eligibility, and inclusion.

Comment 2.2

Reviewer comment

The source count is inconsistent: the abstract reports 185 sources, Section 2 reports 160, and the reference list contains 185 entries of mixed type. Please give the true breakdown by source type and make the counts consistent throughout the manuscript.

Original manuscript

The original manuscript was internally inconsistent: the abstract referred to approximately 185 sources while the literature-review text said “over 160 sources.” The reference list and evidence-base count were not clearly distinguished.

Current revised manuscript

The current highlighted Section 1.6.3 now gives a precise qualitative-synthesis breakdown: **92 peer-reviewed journal articles + 21 conference papers + 14 technical standards/patent documents + 38 government/agency/institutional reports + 20 corporate/grey-literature sources = 185**. This directly resolves the earlier “over 160” wording.

However, the current uploaded bibliography appears to end at reference [178], while the manuscript states that 185 distinct sources were included in the final qualitative synthesis. There is also at least one body citation to [186], which is not present in the current reference list.

Highlighted revision evidence

The source-count paragraph in Section 1.6.3 is highlighted. The source-type donut in Figure 2 also shows the 92/21/14/38/20 composition totaling 185.

Assessment

PARTIALLY ADDRESSED

The conceptual distinction is now correct, but the current file is not yet numerically auditable: a 185-source included corpus should be traceable to the bibliography or to a supplementary evidence table. A bibliography with only 178 numbered entries cannot, without further explanation, support 185 distinct included sources.

Required final action before resubmission

Reconcile the bibliography against the 185-source evidence corpus. Because 185 is the intended final included count, verify which included sources are missing from the current reference list and restore/correct them. Remove or repair the dangling [186] citation. Then report the final bibliography count separately from the 185-source synthesis count.

Recommended response to the reviewer

We apologize for the inconsistency in the original manuscript. The final qualitative synthesis includes **185 sources**, comprising 92 peer-reviewed journal articles, 21 conference papers, 14 standards and patent documents, 38 agency and institutional reports, and 20 corporate and grey-literature sources. The earlier “over 160 sources” wording has been removed. We now distinguish the screened evidence corpus from the complete bibliography, because methodology, definitions, standards, and background references may fall outside the screened qualitative synthesis. After final reference reconciliation, the bibliography contains **[INSERT VERIFIED FINAL BIBLIOGRAPHY COUNT]** numbered entries, while the screened qualitative evidence base remains $n = 185$.

Comment 2.3

Reviewer comment

Sections 5.3, 7.3, 10.1, 10.2, and 10.4 present original modelling results (e.g., the 720 MWh deterministic baseline, the >90% storage-reduction claim, the Pareto-frontier figure, the payback ranges) but no model is documented. Please either document the model in an appendix with a data availability statement, or replace these outputs with sourced published values.

Original manuscript

The original manuscript presented numerical outputs as though they came from an original optimization/modeling exercise, including a 720 MWh deterministic baseline, >90% storage reduction, Pareto-frontier outputs, and generalized payback results, without documenting the model, data, solver, or reproducibility information.

Current revised manuscript

The current revision removes the 720 MWh baseline and the universal >90% storage-reduction claim. Section 5.3 now explicitly states that the review “does not claim a universal storage reduction or a site-specific Pareto-optimal configuration.” Section 7.3 similarly says that the framework does not specify a universal optimum and requires model, data, solver, time resolution, assumptions, and outputs to be documented if an optimization model is used.

The revised manuscript retains generic equations in Section 5.3, but they are now introduced as a **conceptual mathematical structure synthesized from the literature** and explicitly state that they were **not used to generate site-specific numerical results**. This is materially different from the original presentation.

Highlighted revision evidence

Section 5.3 is one of the most heavily highlighted parts of the revised file. Section 7.3 is also extensively highlighted, and the former Pareto-output framing has been replaced by a conceptual technology-selection framework.

Assessment

ADDRESSED

The reviewer’s central concern has been addressed: unsupported

original numerical outputs are no longer presented as results. Retaining generic equations is acceptable if they are clearly labeled as literature-synthesized conceptual formulations rather than executed models.

Required final action before resubmission

Keep the conceptual-formulation disclaimer immediately before the equations. Ensure no figure, table, abstract sentence, or conclusion describes the conceptual equations as a completed optimization or reports an output from them.

Recommended response to the reviewer

We agree with the reviewer. The unsupported original numerical outputs have been removed rather than presented without a reproducible model. The former 720 MWh deterministic baseline, universal storage-reduction percentage, and site-specific Pareto-frontier claims are no longer reported. Section 5.3 now presents only a generic mathematical structure synthesized from the literature and states explicitly that it was not used to generate site-specific numerical results. Section 7.3 likewise requires any future optimization application to document the model, data, solver, time resolution, assumptions, and outputs. The framework is therefore presented as a screening and decision-support structure rather than as an undocumented optimization result.

Comment 2.4

Reviewer comment

LOLE is defined as hours per year, but the equation divides the indicator sum by 8760, returning a probability rather than an expectation in hours. Please either remove the factor of 8760 or

relabel the metric as LOLP (loss-of-load probability).

Original manuscript

The original LOLE equation divided the indicator sum by 8,760, which normalized the result and therefore produced a probability rather than hours/year.

Current revised manuscript

The revised Section 5.3 defines LOLE as the accumulated duration of time steps in which available power is below the critical load. The corrected form is conceptually $LOLE = \sum 1[P_{avail,t} < P_{crit,t}] \Delta t$, so the result has units of hours when Δt is expressed in hours.

Highlighted revision evidence

The entire corrected adequacy discussion in Section 5.3 is highlighted.

Assessment

ADDRESSED

The dimensional problem identified by the reviewer is corrected. The separate issue of how the manuscript relates LOLE to an annual availability target is treated under Comment 2.8 below.

Required final action before resubmission

Retain the no-/8760 formulation and make sure the equation label and variable definitions remain visible in the final Word/PDF export.

Recommended response to the reviewer

We agree. Equation (2) has been rewritten so that LOLE is expressed as the accumulated duration of supply-shortfall intervals rather than as a normalized probability. The factor of $1/8,760$ has been removed. With Δt expressed in hours, the revised summation returns hours over the assessment period, which is dimensionally consistent with LOLE. LOLP is therefore not used as a substitute label for the revised equation.

Comment 2.5

Reviewer comment

Equation (1) uses τ as the duration but the maximum runs over 1 to n , where n is the count of drought events — i.e., the max is over events, while the index inside the sum is a duration index. Please rewrite as a maximum over events, and state whether P_{load} is the full facility load or only the critical non-deferrable load.

Original manuscript

The original deterministic equation mixed event and duration indices and used a generic P_{load} term without stating whether deferrable training/batch demand was included.

Current revised manuscript

The revised Section 5.3 defines a maximum over discrete deficit events and accumulates the positive energy deficit over the time intervals belonging to each event. It also replaces the generic load term with $P_{crit,t}$, defined as the **critical, non-deferrable load**. The surrounding framework text distinguishes critical services from training, analytics, and batch workloads that may be deferred where technically and contractually feasible.

Highlighted revision evidence

The deterministic equation definitions and the critical-load definitions are highlighted in Section 5.3; related workload-flexibility constraints are highlighted in Sections 7.6 and 11.

Assessment

ADDRESSED

This directly addresses both parts of the reviewer comment.

Required final action before resubmission

Check the final equation rendering because some variable symbols are missing in the plain-text extraction; ensure i , T_i , $P_{crit,t}$, $P_{supply,t}$, and Δt are displayed correctly in the submitted PDF.

Recommended response to the reviewer

We agree. The deterministic sizing equation is now written as a maximum over discrete renewable-deficit events, with the energy deficit summed over the time intervals associated with each event. We also replaced the generic P_{load} term with $P_{crit,t}$, which is explicitly defined as the critical, non-deferrable facility load. Deferrable training, analytics, and batch-processing loads are excluded from the critical-load definition only when deferral is technically feasible and permitted by the applicable service-level and operational requirements.

Comment 2.6

Reviewer comment

Section 5.3 claims that LOLE-based sizing cuts storage by more than 90% against a 720 MWh baseline, but the baseline is never derived and the facility size is never stated. Section 13.3 separately gives 30–60% for the same comparison. Please reconcile.

Original manuscript

The original manuscript simultaneously reported >90% storage reduction relative to an unexplained 720 MWh baseline and a separate 30–60% reduction, creating a direct contradiction.

Current revised manuscript

Both universal reduction claims have been removed from the current manuscript. Section 5.3 now says that no universal storage reduction is claimed. The revised discussion instead requires chronological, site-specific renewable-resource, load, grid, and backup data to quantify any reduction.

Highlighted revision evidence

The corrective disclaimer in Section 5.3 is highlighted; related cautionary language appears in the highlighted framework and conclusion text.

Assessment

ADDRESSED

The contradiction itself has been removed rather than numerically “reconciled,” which is the stronger response because the original values were not reproducibly derived.

Required final action before resubmission

Verify that no residual 30–60%, 40–60%, 50–70%, or >90% **storage-reduction** statement remains elsewhere unless it is clearly tied to a specific cited study and system boundary. Percentages used for unrelated metrics should not be removed merely because the same number appears.

Recommended response to the reviewer

We agree. The two storage-reduction claims were not supported by a common reproducible baseline, so both have been removed. The revised manuscript no longer reports either the >90% reduction against a 720 MWh baseline or the 30–60% alternative range as a general result. Section 5.3 now states explicitly that this review does not claim a universal storage reduction, and any comparison between deterministic and probabilistic sizing must be quantified using chronological, site-specific load, renewable-resource, grid-availability, and operational data.

Comment 2.7

Reviewer comment

Section 7.3 expresses battery-cost sensitivity in \$/MWh. Battery capital cost is a \$/kWh quantity. Please correct the unit and state whether the price refers to the cell, the pack, or the installed system.

Original manuscript

The original Section 7.3 expressed battery-cost sensitivity in USD/MWh, conflating an energy-service metric with battery capital cost and failing to define whether the cost referred to cells, packs, or a fully installed BESS.

Current revised manuscript

The current manuscript now reports the BNEF 2025 **battery-pack** benchmark as USD 108/kWh and explicitly explains that a fully installed BESS also includes PCS, thermal management, controls, fire protection, EPC, site development, and grid interconnection. Table 3 labels entries as “Battery-pack price” or “Installed-system cost.” Section 10.1 separately reports an installed-system benchmark.

Highlighted revision evidence

Table 3 is fully highlighted in the current Word file, and the pack-vs-installed distinction is also highlighted in Sections 7.8 and 10.1.

Assessment

ADDRESSED

The unit and system-boundary problem is addressed.

Required final action before resubmission

Use USD/kWh for capital-cost metrics and reserve USD/MWh for LCOE/LCOS or energy-service metrics. Keep the words “battery-pack price” and “installed BESS cost” adjacent to the values.

Recommended response to the reviewer

We agree. Battery capital cost is now reported in USD/kWh and the system boundary is explicitly identified. The BNEF USD 108/kWh value is labeled as a 2025 volume-weighted **battery-pack** price, not a fully installed BESS cost. Installed-system costs are discussed separately and include the power-conversion system, thermal management, controls, fire protection, engineering, procurement, construction, site development, and interconnection. Table 3 now labels pack and installed-system values separately, while LCOS

values remain in USD/MWh of storage service.

Comment 2.8

Reviewer comment

LOLE of 0.1 h/yr corresponds to 99.9989% availability, not 99.99% as stated. The reliability target also switches between 99.9% and 99.99%. Please fix the arithmetic and settle on a single reliability target.

Original manuscript

The original manuscript incorrectly stated that 0.1 h/year corresponded to 99.99% availability and switched between 99.9% and 99.99% targets.

Current revised manuscript

The revised manuscript removes the 0.1 h/year value and uses the arithmetic pair **99.99% annual availability** ↔ **0.876 h/year of annual unavailability** under a simplified duration calculation. This arithmetic is correct: $8,760 \times (1 - 0.9999) = 0.876$ h/year.

However, the current text still phrases this as though a 99.99% availability target *gives a LOLE of 0.876 h/year*. That is conceptually unsafe. Availability/unavailability and LOLE are different reliability/adequacy metrics and should not be presented as mathematically interchangeable. Also, the reviewer's original arithmetic was correct: 0.1 h/year of unavailability corresponds to approximately **99.9989%** availability under the same simple annual-duration relation.

Highlighted revision evidence

The reliability wording is highlighted in Section 5.3 and appears again in the framework reliability discussion.

Assessment

PARTIALLY ADDRESSED — IMPORTANT

The arithmetic inconsistency is fixed, but the metric-definition issue is only partially fixed. The older response letter should **not** challenge the reviewer on the 99.9989% calculation; the reviewer was correct.

Required final action before resubmission

Replace the current wording with: “An illustrative annual availability target of 99.99% corresponds to an annual unavailability allowance of 0.876 h/year under a simple duration calculation. This allowance is not itself LOLE. LOLE is calculated separately from the chronological adequacy model in Eq. (2). If an illustrative LOLE threshold is adopted for screening, it should be stated as an independent planning criterion rather than derived from availability.” Also avoid using $LOLE \leq 0.876$ h/year unless the manuscript explicitly adopts that as an independent screening assumption.

Recommended response to the reviewer

We thank the reviewer for identifying the inconsistency. The reviewer is correct that 0.1 h/year corresponds to approximately 99.9989% availability under a simple annual-duration calculation. The 0.1 h/year statement has therefore been removed. The revised manuscript distinguishes the two concepts: 99.99% annual availability corresponds to 0.876 h/year of annual unavailability under a simple duration calculation, whereas LOLE is calculated independently from chronological adequacy simulation using Equation (2). We no longer present availability and LOLE as mathematically interchangeable metrics. Any LOLE threshold used for screening is identified as a separate planning assumption and must be validated for the project-specific reliability requirement.

Comment 2.9

Reviewer comment

Table 7 justifies the grid-only LCOE using PJM/CAISO/ERCOT capacity prices. Capacity prices clear in \$/MW-day, not \$/MWh, and a capacity price is not an LCOE. Please separate the energy, capacity, ancillary, and delivery components and cite a tariff or market report for each.

Original manuscript

The original Table 7 treated regional capacity-market values as “grid-only LCOE,” mixing USD/MW-day capacity prices with USD/MWh energy costs and implying that a capacity price was an LCOE.

Current revised manuscript

The revised economic section now separates energy and capacity components. Table 7 classifies grid electricity as a **Firm Delivered Service** with a **location-specific** cost rather than a universal LCOE. Section 10.2 explains that wholesale energy is commonly reported in USD/MWh while capacity may be USD/MW-day and includes an illustrative capacity-cost conversion. It also lists transmission, distribution, demand, ancillary-service, tax, and tariff components needed for an all-in grid cost.

Highlighted revision evidence

The revisions to Table 7, Table 9, and most of Section 10.2 are highlighted.

Assessment

ADDRESSED

The reviewer's conceptual objection is addressed. The revised manuscript no longer assigns universal PJM/CAISO/ERCOT "grid-only LCOEs."

Required final action before resubmission

Use USD consistently instead of alternating \$ and USD. Keep the PJM conversion explicitly labeled as an **illustrative capacity-cost allocation**, not a delivered tariff or LCOE.

Recommended response to the reviewer

We agree. The previous "grid-only LCOE" rows have been removed. The revised manuscript now treats capacity and energy as separate market components and identifies capacity prices in USD/MW-day rather than USD/MWh. Section 10.2 shows an illustrative conversion only to demonstrate how a capacity price may be allocated over energy consumption for a specified load factor; it is not presented as an LCOE or full delivered tariff. Grid electricity is now reported as location- and tariff-specific and must include the applicable energy, capacity (where relevant), ancillary-service, transmission, distribution, demand, tax, and utility-fee components.

Comment 2.10

Reviewer comment

Table 7 places unfirmed solar/wind LCOE alongside firm grid supply; the headline conclusion rests on that comparison. Please either add

firming cost and curtailment to the renewable rows, or limit the comparison to firmed hybrid rows.

Original manuscript

The original table placed generation-only solar and wind LCOEs beside firm grid supply and used the apparent difference to support a headline economic conclusion.

Current revised manuscript

The current Table 7 explicitly labels solar and wind as **Unfirmed Generation**, U.S. solar-plus-storage and Australian renewable-storage portfolios as **Partially Firmed Hybrid**, BESS as **Storage Service**, and grid electricity as **Firm Delivered Service**. Table 9 repeats the distinction. Section 10.2 now says generation-only LCOE should not be directly compared with all-in firm delivered-grid service.

Highlighted revision evidence

Table 7 and the Section 10.2 cost-boundary discussion are highlighted; Figure 10 is also redesigned around the firmed/unfirmed distinction and explicitly avoids a universal grid-cost bar.

Assessment

ADDRESSED

This is one of the strongest corrections in the revision.

Required final action before resubmission

Rewrite the poorly worded highlighted paragraph beginning “Unfirmed generation cost values...” using the cleaner cost-boundary wording from the grammar audit. The technical logic is correct; the prose is not.

Recommended response to the reviewer

We agree. The revised manuscript no longer treats generation-only renewable LCOE and firm delivered-grid electricity as equivalent products. Tables 7 and 9 now classify each benchmark by system boundary. Standalone solar and wind are labeled as unfirmed generation and explicitly exclude storage, backup, curtailment, transmission, interconnection upgrades, and reliability provisions. Renewable-storage portfolios are labeled as partially firmed hybrids, while grid electricity is treated as a location- and tariff-specific firm delivered service. The conclusions have also been rewritten so that no general cost advantage is inferred from a direct comparison between unfirmed generation and firm supply.

Comment 2.11

Reviewer comment

Table 7 assumes a 5–7% real discount rate, Section 7.8 applies 7.5%, and Section 10.4 uses 7–12%. Please state a single WACC, indicate whether it is real or nominal, and apply it to every LCOE, NPV, IRR, and payback calculation.

Original manuscript

The original manuscript used multiple discount-rate assumptions: 5–7%, 7.5%, and 7–12%, without a consistent distinction between real and nominal rates.

Current revised manuscript

The current revision uses **7.5% real WACC** as an illustrative screening

assumption and **5–12% real** as a sensitivity range. It also states that real cash flows should be discounted with real rates and nominal cash flows with nominal rates plus an explicit inflation assumption.

Importantly, the current paper now relies heavily on **published LCOE benchmarks** rather than recalculating every source using the paper's WACC. Published IRENA/Lazard/other LCOEs retain their own source assumptions and should not be described as if they were all recomputed at 7.5%.

Highlighted revision evidence

Sections 7.8 and 10.4 are heavily highlighted.

Assessment

ADDRESSED WITH CLARIFICATION

The internal financing convention is now consistent, but the response should clarify that 7.5% applies to the manuscript's illustrative project-finance screening, not retroactively to externally published LCOEs.

Required final action before resubmission

Keep the 7.5%/5–12% convention only where the manuscript performs or illustrates its own financial calculations. Label externally sourced LCOEs as literature benchmarks with source-specific financing assumptions.

Recommended response to the reviewer

We agree that the original manuscript used inconsistent discount-rate assumptions. The revised manuscript adopts a **7.5% real WACC** as a single illustrative screening assumption for project-specific financial analysis, with a **5–12% real range** used only for sensitivity analysis. Real and nominal cash flows are now distinguished explicitly. The

literature-based LCOE values reported from IRENA, Lazard, and other sources are not recalculated using the manuscript's 7.5% assumption; they are retained as source-reported benchmarks and are labeled accordingly. No universal project-specific WACC is recommended.

Comment 2.12

Reviewer comment

Payback is variously reported as 5–12 yr, then 6–11 yr, then 4–8 yr; LCOE as 35–100 and then 35–80 \$/MWh; storage reduction as 40–60% and then 50–70%; interconnection cost as USD 50k–500k plus USD 10M (Section 1.3) vs USD 500k–2M or USD 5–10M (Section 7.2). Please build a single results table and reconcile all headline numbers.

Original manuscript

The original manuscript reported mutually inconsistent payback, LCOE, storage-reduction, and interconnection-cost ranges in different sections.

Current revised manuscript

The current revision removes the universal payback ranges and storage-reduction percentages. The abstract/conclusion no longer relies on a single LCOE range. Tables 7 and 9 organize cost benchmarks by cost boundary rather than forcing a single “result.” The interconnection text now uses the Section 1.3 queue/study/upgrade example rather than maintaining the competing Section 7.2 ranges.

Highlighted revision evidence

Large portions of Sections 7.8 and 10.1–10.5 are highlighted, as are the revised Table 7 and Table 9 structures.

Assessment

SUBSTANTIALLY ADDRESSED

The main inconsistency has been resolved by removing unsupported generalized results and replacing them with source-specific benchmarks. This is more defensible than constructing a false single result table from incomparable values.

Required final action before resubmission

Run one final numerical-search pass for all old headline values before submission. Also correct the residual cost inconsistency noted under Comment 2.24 and the dangling [186] citation.

Recommended response to the reviewer

We agree. The conflicting headline ranges in the original manuscript have been removed rather than averaged or combined. The revised manuscript no longer reports a universal payback range or a universal storage-reduction percentage. LCOE and hybrid-system benchmarks are now organized by system boundary and source in Tables 7 and 9, and project-specific NPV, IRR, and payback must be calculated from auditable cash flows. The interconnection discussion has also been consolidated so that study fees, queue time, and potential network-upgrade costs are not presented as interchangeable cost categories. The conclusions are qualitative and no longer depend on the inconsistent headline numbers identified by the reviewer.

Comment 2.13

Reviewer comment

The abstract calls USD 108/kWh a 2026 projection, while Sections 1.2 and 2.3 correctly report it as an observed 2025 BNEF figure. The volume-weighted pack price is also never reconciled with the USD 75–160/kWh installed-system costs quoted later.

Original manuscript

The original abstract described USD 108/kWh as a 2026 projection and later mixed the pack-level value with installed-system cost ranges.

Current revised manuscript

The current manuscript describes USD 108/kWh as the **2025 BNEF volume-weighted battery-pack price** and explicitly explains that the value excludes PCS, thermal management, controls, EPC, construction, site development, fire protection, and interconnection. Section 10.1 separately reports an installed BESS benchmark, and Table 3 labels pack and installed-system values.

Highlighted revision evidence

The pack-vs-installed distinction is highlighted in Section 10.1 and Table 3.

Assessment

ADDRESSED

The specific temporal and system-boundary issue is addressed.

Required final action before resubmission

Use USD 108/kWh consistently, and verify the BNEF source is the final

authoritative citation in the bibliography.

Recommended response to the reviewer

We agree. The USD 108/kWh value is now consistently identified as a **2025 observed volume-weighted lithium-ion battery-pack price**, not a 2026 projection and not a fully installed BESS cost. The manuscript now separates battery-pack price from installed-system cost and explicitly lists the additional balance-of-system, power-conversion, thermal-management, control, engineering, construction, site-development, fire-protection, and grid-interconnection costs that are outside the pack price.

Comment 2.14

Reviewer comment

(a) Section 1.1 attributes '2.9 Wh per GPT-3 query' to 'research by Li', but reference [7] is Shi/Yang/Lo and no Li source appears. (b) Reference [58] is BloombergNEF 'Corporate Clean Energy Buying Fell in 2025', cited in Section 10.2 and Table 3 for BNEF battery pack prices. (c) Reference [150] is Lazard June 2025, but Table 7 states that values reflect 2026 market conditions validated against NREL/BNEF/IRENA, citing [150] for all of it.

Original manuscript

The original manuscript used three citations incorrectly: ref. [7] for a 2.9 Wh/query claim attributed to "Li"; ref. [58] for BNEF pack prices even though it was a different BNEF publication; and a Lazard reference as if it substantiated all market benchmarks.

Current revised manuscript

The current manuscript removes the 2.9 Wh/query number and uses ref. [7] only for a general statement that inference energy can become material at scale. The BNEF pack price is now tied to ref. [17]. Cost tables now identify IRENA/Lazard/other sources by the specific benchmark rather than using one citation to support all values.

Highlighted revision evidence

The relevant corrected passages occur in the revised introduction, Table 3, Table 7, and Section 10.1; many of those revised economic passages are highlighted.

Assessment

ADDRESSED FOR THE THREE NAMED CITATIONS

The three reviewer-identified citation mismatches are materially corrected. A broader citation audit is still warranted because the current file contains a dangling [186] citation and other bibliography-format problems.

Required final action before resubmission

Verify every precise numerical claim against its source after the final reference renumbering. Do not assume the reference numbers in the older response letter remain valid.

Recommended response to the reviewer

We thank the reviewer for identifying these citation mismatches. The unsupported 2.9 Wh/query claim and the incorrect attribution to “Li” have been removed. The 2025 BNEF battery-pack price is now supported by the dedicated battery-price source, and the economic tables no longer use a single Lazard citation to stand in for NREL,

BNEF, IRENA, or other independent benchmarks. We also rechecked the surrounding numerical claims against their source-specific system boundaries and years.

Comment 2.15

Reviewer comment

(a) Section 10.5 attributes downtime >USD 500,000/h to ref [9] (Uptime Institute survey), which reports outage-cost distributions rather than a per-hour figure for training clusters. (b) Section 10.3 quotes a PPA price of USD 64.49/MWh to two decimals with no citation. (c) Section 1.1 gives 165% demand growth by 2030 with no baseline year. (d) Section 8.1 gives Google figures with no reporting year.

Original manuscript

The original manuscript reported a universal >USD 500,000/h downtime cost, an uncited PPA value, a 165% growth figure without a baseline year, and Google metrics without a reporting year.

Current revised manuscript

The current manuscript removes the >USD 500,000/h universal downtime claim and instead says Uptime Institute reports outage-cost distributions. The LevelTen PPA figures now carry a citation. The Goldman Sachs growth statement now specifies **2024 to 2030**. Google's hourly CFE paragraph now states **2024** as the reporting year.

Highlighted revision evidence

The revised PPA section and business-case/critical-appraisal section are highlighted; the Google case-study revisions are also highlighted.

Assessment

SUBSTANTIALLY ADDRESSED

The four factual support issues are substantially addressed. One manuscript-style problem remains: Section 10.5 contains the revision-history sentence “The previous per-hour downtime value has been thus eliminated.” That sentence belongs in the response letter, not in the paper.

Required final action before resubmission

Delete the revision-history sentence from Section 10.5. Retain only the corrected neutral statement about outage-cost distributions and project-specific downtime valuation.

Recommended response to the reviewer

We agree. The unsupported universal downtime-cost figure has been removed. The manuscript now refers only to the outage-cost distributions reported by Uptime Institute and states that any project downtime cost must be derived from affected IT capacity, workload value, recovery time, contractual penalties, and lost utilization. The Q1 2026 PPA values are now cited to the LevelTen market report; the 165% demand-growth estimate is explicitly stated as a 2024–2030 change; and the Google energy/CFE figures now identify the reporting year.

Comment 2.16

Reviewer comment

Table 1 asserts gaps in four families of prior work but contains no citations in any cell. Please add representative references.

Original manuscript

The original Table 1 asserted technical, economic, case-study, and equipment-centric research gaps without citations in the individual cells.

Current revised manuscript

The current Table 1 contains citations throughout its technical-strength, blind-spot, and barrier cells. The entire table is highlighted, indicating a direct revision in response to the reviewer.

Highlighted revision evidence

Table 1 is fully highlighted in the current Word file.

Assessment

ADDRESSED

The reviewer's specific request is addressed.

Required final action before resubmission

During final reference renumbering, verify every in-cell citation still maps to the source it is intended to support.

Recommended response to the reviewer

We agree. Representative references have been added throughout

Table 1. Each of the four prior-work families now includes citations supporting the stated technical strengths, economic blind spots, and implementation barriers. The table is therefore no longer an uncited assertion of research gaps; it is a referenced synthesis of the literature that motivates the scope of the seven-component framework.

Comment 2.17

Reviewer comment

Components 2, 3, and 4 of the framework are described only in general terms. Please state the decision rules, the inputs each component requires, and the output it produces.

Original manuscript

The original framework described Components 2–4 largely at a conceptual level and did not consistently specify required data, selection rules, and outputs.

Current revised manuscript

The revised Sections **7.3, 7.4, and 7.5** are among the most heavily highlighted sections. Section 7.3 now states the inputs required for technology selection, rejects configurations that violate reliability/interconnection/power/energy/emissions constraints, and defines outputs such as shortlisted renewable capacity, storage duration/capacity, grid limits, backup resources, curtailment, cost, and implementation risks. Section 7.4 gives electrical-architecture inputs and design outputs, while Section 7.5 describes primary/secondary/tertiary control, required forecasts/state variables, and dispatch outputs.

Highlighted revision evidence

Section 7.3 has 27 highlighted body paragraphs, Section 7.4 has 22, and Section 7.5 has 20, making this one of the clearest reviewer-driven revisions.

Assessment

ADDRESSED

The reviewer's request for decision rules, inputs, and outputs is addressed in substance.

Required final action before resubmission

Keep the input/rule/output organization explicit, preferably with consistent subheadings or a compact table if journal space allows.

Recommended response to the reviewer

We agree that the original framework was too general. Components 2, 3, and 4 have been expanded so that each now states the data required, the governing decision or screening rule, and the output produced. Component 2 defines the resource, load, cost, storage, tariff, interconnection, backup, reliability, and emissions inputs and produces a shortlist of feasible technology configurations. Component 3 converts the selected configuration into an electrical architecture using load, harmonic, grid-code, equipment, and protection inputs. Component 4 specifies the hierarchical control levels, required forecasts and state variables, and the resulting active/reactive-power, curtailment, storage, and workload-control signals.

Comment 2.18

Reviewer comment

Section 7.2 recommends MCDA but gives no criteria, no weights, no weighting method (AHP or entropy), and no sensitivity check. As written, it is a label rather than a method.

Original manuscript

The original Section 7.2 mentioned MCDA but did not provide a reproducible weighting method, explicit weights, a consistency test, or a defined sensitivity procedure.

Current revised manuscript

The current highlighted Section 7.2 now specifies **AHP**, five criteria, illustrative weights of 0.30/0.25/0.20/0.15/0.10, normalized site scores, and an AHP consistency-ratio target below 0.10. Table 5 provides the criteria and weights.

The remaining weakness is the sensitivity sentence. It only says that rankings should be repeated by “changing the weightings assigned to the principle criterion” and checking whether rankings move. It does not define the perturbation magnitude or renormalization rule, so the sensitivity test is not fully reproducible.

Highlighted revision evidence

Section 7.2 is highlighted, although Table 5 itself is not consistently highlighted at cell level.

Assessment

PARTIALLY ADDRESSED

The MCDA is no longer merely a label, but the sensitivity step should be specified numerically.

Required final action before resubmission

Replace the current sensitivity sentence with a reproducible rule, for example: “A one-at-a-time sensitivity analysis is performed by perturbing each baseline criterion weight by $\pm 20\%$ relative to its baseline value and proportionally renormalizing the remaining weights so that the total remains 1.0. Rank stability of the top three candidate sites is then recorded.” Keep the weights explicitly illustrative rather than universal.

Recommended response to the reviewer

We agree that the original MCDA description was insufficient. The revised Section 7.2 now defines an AHP-based procedure, identifies five site-selection criteria, provides illustrative baseline weights, normalizes candidate-site scores, and requires an AHP consistency ratio below 0.10. We have also added a reproducible one-at-a-time sensitivity test in which each baseline weight is perturbed by $\pm 20\%$ relative to its baseline value and the remaining weights are proportionally renormalized to preserve a total weight of 1.0. The top-three ranking is then checked for stability. The weights are explicitly presented as illustrative screening values that must be recalibrated using project-specific stakeholder judgments.

Comment 2.19

Reviewer comment

Section 6 is a catalogue of tool categories. Please state which modelling approach suits which question, at what time resolution, with what data requirements, and where each approach breaks down.

Original manuscript

The original Section 6 read primarily as a list of software and modeling categories rather than guidance on which method should be used for which engineering question.

Current revised manuscript

The revised manuscript adds Table 4, which maps eight engineering questions to recommended methods, typical time resolution, required inputs, and principal limitations. It distinguishes steady-state AC/DC power flow, harmonic analysis, EMT simulation, chronological time-series simulation, lifecycle/MILP analysis, MPC, and workload scheduling. The text also explains the role of HOMER, MATLAB/Simulink, PSCAD/EMTDC, pandapower/PYPOWER, and related tools.

Highlighted revision evidence

The new Table 4 is not heavily highlighted at cell level, but the surrounding Section 6 revision and the modeling-method discussion are clearly expanded relative to the original.

Assessment

ADDRESSED

The reviewer's requested method-question-time-resolution-data-limitation mapping is addressed.

Required final action before resubmission

Use U.S. spelling consistently (*modeling*, not *modelling*) if that is the selected manuscript style.

Recommended response to the reviewer

We agree. Section 6 has been revised from a catalogue of tools into a

question-oriented modeling comparison. Table 4 now states, for each major engineering question, the recommended analysis method, typical time resolution, required input data, and principal limitation. The revised discussion distinguishes steady-state power flow, frequency-domain harmonic analysis, EMT simulation, chronological time-series simulation, lifecycle/MILP analysis, MPC, and workload-scheduling methods so that the modeling approach is tied to the question being asked rather than to software availability alone.

Comment 2.20

Reviewer comment

Water use effectiveness and the water-vs-energy tradeoff in cooling selection are treated only as a siting input, despite the sustainability framing. Please add a dedicated discussion.

Original manuscript

The original sustainability discussion mentioned water mainly as a siting consideration and did not adequately discuss the operational tradeoff between cooling electricity and water consumption.

Current revised manuscript

The current manuscript adds dedicated discussion of Water Usage Effectiveness (WUE), direct versus indirect water consumption, cooling-system water/energy tradeoffs, water-stress exposure, and the need to evaluate PUE and WUE together. Section 12.4 further explains that air-cooled systems may reduce direct water consumption while increasing cooling electricity, whereas evaporative/liquid configurations can reduce cooling energy while increasing water demand.

Highlighted revision evidence

The environmental-impact paragraph in Section 12.4 is highlighted, and the cooling-coordination discussion in Section 7.7 is also expanded.

Assessment

ADDRESSED

The reviewer's requested sustainability tradeoff is now explicitly discussed.

Required final action before resubmission

Avoid unsupported universal percentage claims for air-vs-evaporative cooling unless the cited source and system boundary support them. The current qualitative wording is safer.

Recommended response to the reviewer

We agree. The revised manuscript now treats water use as an operational sustainability tradeoff rather than only a siting input. The cooling discussion distinguishes direct on-site water use from indirect water use embedded in electricity supply, defines WUE, and explains that cooling architectures can trade lower electrical demand against higher water consumption. The framework therefore requires cooling alternatives to be assessed using both energy- and water-related metrics and to account for local water stress and permitting conditions.

Comment 2.21

Reviewer comment

Section 11 presents workload deferral as a purely technical option, without acknowledging commercial reluctance to curtail training jobs on assets of this capital cost.

Original manuscript

The original workload-flexibility discussion largely treated training and batch jobs as technically deferrable without acknowledging the commercial value of GPU utilization, deadlines, contracts, and SLAs.

Current revised manuscript

The current highlighted Section 7.6 explicitly states that workload deferral is constrained by technical **and commercial** requirements. It identifies model-development schedules, GPU utilization targets, contractual commitments, SLAs, checkpointing, data residency, cybersecurity, network latency, and business objectives. Table 6 also labels flexibility as potential rather than guaranteed.

Highlighted revision evidence

The workload-deferral paragraph in Section 7.6 is highlighted, and multiple related workload-balancing paragraphs in Section 11 are highlighted.

Assessment

ADDRESSED

This directly addresses the reviewer's concern.

Required final action before resubmission

Retain the phrase that technical deferral potential is an upper bound rather than guaranteed dispatchable flexibility; it is an important critical-review qualification.

Recommended response to the reviewer

We agree. The revised manuscript no longer treats workload deferral as a purely technical or freely dispatchable resource. Sections 7.6 and 11 now state that practical flexibility is constrained by GPU-asset utilization, model-development schedules, contractual commitments, SLAs, checkpointing capability, data-residency and cybersecurity requirements, network latency, and business priorities. Table 6 therefore describes potential flexibility rather than guaranteed dispatchable flexibility.

Comment 2.22

Reviewer comment

Please state the boundaries of the review: U.S. bias of case studies, reliance on self-reported corporate sustainability disclosures, and small number of commercial sources behind cost data.

Original manuscript

The original manuscript had no dedicated limitations section; it moved from sustainability/regulation directly to conclusions.

Current revised manuscript

The revised manuscript adds a new Section 13 with five subsections: evidence/geographic scope, corporate reporting/source quality,

methodological/screening limitations, site-specific validation, and technology/policy/market uncertainty. The new text directly acknowledges North American/U.S. bias, self-reported corporate disclosures, source heterogeneity, concentration of commercial cost data, lack of end-to-end facility validation, and rapidly changing market conditions.

Highlighted revision evidence

Sections 13.1–13.5 are highlighted, making the new limitations section one of the clearest additions relative to the original.

Assessment

SUBSTANTIALLY ADDRESSED — LANGUAGE CLEANUP NEEDED

The substantive reviewer request is met. The section still needs language cleanup: the heading remains “13. Limitation,” Section 13.4 is sentence-style with a period, and several sentences are grammatically poor or accidentally reverse the intended meaning.

Required final action before resubmission

Apply the complete language revision in the companion grammar audit. In particular, change the heading to 13. *Limitations*; rewrite 13.2 to state that the review **does not independently audit** corporate data; replace “way broader,” “bench marked,” and the unexplained “SFA”; and retitle 13.4 *Lack of Site-Specific Validation*.

Recommended response to the reviewer

We agree. A dedicated Section 13, “Limitations,” has been added with five subsections. The revised section explicitly addresses the geographic concentration of the evidence base, dependence on self-reported corporate disclosures, heterogeneity among peer-reviewed and grey-literature sources, concentration of some cost benchmarks

in a small number of commercial organizations, absence of end-to-end validation against a single operating facility, and technology/policy/market uncertainty. The section also states that the numerical values in the review are screening benchmarks rather than directly transferable project estimates and that site-specific application requires local technical, regulatory, tariff, reliability, and financing data.

Comment 2.23

Reviewer comment

Of nine figures, only Figure 1 carries a source line. Every caption needs one. Figure 7 is a modelling output and needs its formulation, decision variables, and solver stated. Figure 4 needs site, data source, and years averaged.

Original manuscript

In the original manuscript, only Figure 1 had an explicit source line. Other figure and table captions generally lacked source attribution. The original Pareto-frontier figure was presented as a model output without enough reproducibility information, and the original wind/solar figure lacked sufficient data provenance.

Current revised manuscript

The current manuscript has improved figure/table descriptions, but the reviewer's caption requirement is **not yet fully implemented**. In the current body text, Figure 1 has an explicit source line and Figure 10 has an explicit source line. Most other figure captions — including Figures 2–9 — do not contain an explicit `Source:` line in the caption text. The table captions similarly generally appear as titles without explicit source

lines.

The old Pareto-frontier output has been removed/replaced by conceptual framework figures, which resolves the solver/formulation issue for that figure. However, conceptual figures still need a source line such as “Source: Authors’ synthesis.”

Highlighted revision evidence

Several figures and tables are newly introduced or revised, but the lack of caption-level `Source:` text is visible in the current file. The revised Figure 10 source line is a good model to follow.

Assessment

NOT YET FULLY ADDRESSED

This is a significant remaining reviewer issue and should be fixed before resubmission. Embedded citations inside a figure or citations in nearby body text are not substitutes for a source line in the caption when the reviewer explicitly requested one.

Required final action before resubmission

Add an explicit source line to **every figure and every table caption**. Examples: Figure 2 — “Source: PRISMA 2020 flow-diagram structure; screening counts from this review.” Figure 3 — “Source: Authors’ illustrative composite based on the workload/equipment literature cited in Section 3.2; not a measured facility.” Conceptual framework figures — “Source: Authors’ synthesis.” Data figures — cite the actual IEA/IRENA/NREL/BNEF/etc. source and specify site/year/data period where applicable. Tables 1–10 should each include “Source: Authors’ synthesis based on ...” or the relevant primary sources.

Recommended response to the reviewer

We agree. Every figure and table caption has now been revised to include an explicit source line. Data-derived figures identify the underlying publisher/data source and the applicable year or data period, while conceptual figures are labeled “Source: Authors’ synthesis.” The former Pareto-frontier modeling output has been removed, so no undocumented solver-generated figure remains. The renewable-resource figure caption also states the relevant data source and temporal/geographic basis. These changes make the provenance of every visual element explicit.

Comment 2.24

Reviewer comment

Three sets of capital cost / LCOE figures conflict: Table 5 (original) solar 900–1600 \$/kW + 50–80 \$/MWh, wind 1800–2600 \$/kW + 45–70 \$/MWh, batteries 180–320 \$/kWh; Section 10.1 (original) solar 700–1200 \$/kW, wind 1150–1800 \$/kW, batteries 100–150 \$/kWh; Table 7 (original) solar 25–45 \$/MWh, wind 20–40 \$/MWh. Please reconcile into one set. Table 3 separately mixes pack prices and installed system prices in a single column without labelling.

Original manuscript

The original manuscript used incompatible capital-cost and LCOE values across tables and sections and mixed battery-pack and installed-system costs without labeling the boundary.

Current revised manuscript

The current revision is much better organized. Section 10.1 uses IRENA 2024 global weighted-average installed costs for solar, wind, and BESS; BNEF 2025 for pack prices; and NREL ATB for U.S.-specific project

assumptions. Table 3 labels pack versus installed-system values. Tables 7 and 9 distinguish unfirmed generation, partially firmed hybrids, storage service, and delivered grid service.

However, the current Section 10.1 cites [186] for a sentence even though the current reference list appears to end at [178]. Section 14.1 also retains a broad \$0.70–\$1.20/W solar range that is not explicitly reconciled with the IRENA 2024 USD 691/kW global average in Section 10.1. Different values can legitimately coexist if they use different geographies/system boundaries, but those boundaries must be explicit and correctly sourced.

Highlighted revision evidence

Section 10.1, Table 3, Table 7, and Table 9 contain substantial highlighted revisions.

Assessment

PARTIALLY ADDRESSED

The cost-boundary problem is largely solved, but the final reference/cost reconciliation is incomplete.

Required final action before resubmission

Repair the dangling [186] citation; either source or remove the Section 14.1 solar range; verify every Table 3 installed-system range against its cited source and year; and use USD consistently. Do not force one universal number across global IRENA averages and U.S. NREL estimates — instead, label the geography, year, and system boundary.

Recommended response to the reviewer

We agree. The conflicting values in the original manuscript have been reorganized around a consistent set of source-specific cost

boundaries. IRENA is used for global weighted-average commissioned-project costs and LCOEs, BNEF for battery-pack prices, NREL ATB for U.S.-specific technology assumptions, and other sources only where their system boundary is explicitly identified. Table 3 now distinguishes pack prices from installed-system costs, while Tables 7 and 9 separate unfirmed generation, storage service, partially firmed hybrid supply, and delivered-grid electricity. We have also completed a final reference reconciliation so that every numerical benchmark is linked to the correct source, geography, year, and system boundary.

Comment 2.25

Reviewer comment

Section 7 opens with the corrupted phrase 'workload flexModelling economic viability', suggesting find-and-replace damage elsewhere. Please check the whole file.

Original manuscript

The original Section 7 contained the corrupted string “workload flexModelling economic viability,” indicating find-and-replace damage.

Current revised manuscript

The current Section 7 opening no longer contains that corrupted phrase. It separately lists workload flexibility modeling and economic viability.

Highlighted revision evidence

The surrounding framework section has been extensively revised,

although the opening paragraph itself is not highlighted in the current file.

Assessment

ADDRESSED

The specific corruption is fixed.

Required final action before resubmission

Perform one final document-wide search for mid-word capitalization, duplicated words, missing spaces, and broken formula variables. The grammar audit identifies several remaining corruption-like strings elsewhere.

Recommended response to the reviewer

We agree. The corrupted phrase has been removed and Section 7 now treats workload-flexibility modeling and economic viability as separate framework components. We also performed a document-wide search for similar find-and-replace artifacts and corrected any remaining merged words, duplicated fragments, or broken spacing before final submission.

Comment 2.26

Reviewer comment

Section 3.4 reads 'Colling System Integration', Section 3.2 reads 'Al Workload Electrical Charactersitics', Section 1.1 has 'data canters', and Section 1.2 has a stray dollar sign in '\$THD < 5%'. Please proofread the whole file.

Original manuscript

The original manuscript contained the reviewer-identified errors
Colling System Integration, AI Workload Electrical
Charactersitics, data canthers, and \$THD < 5%.

Current revised manuscript

The current uploaded revised manuscript still contains **three of the four exact reviewer-identified errors**:

- Section 1.1 still says “**Data canthers**”.
- Section 3.2 is still titled “**AI Workload Electrical Charactersitics.**”
- Section 3.4 is still titled “**Colling System Integration.**” The stray \$ before THD appears to have been removed.

Highlighted revision evidence

These three remaining errors are in non-highlighted text, which may explain why they survived a revision focused primarily on highlighted additions.

Assessment

NOT ADDRESSED IN THE CURRENT UPLOAD

This reviewer comment is **not closed** in the current upload. The older response letter says these typos were fixed, but the actual current manuscript contradicts that claim.

Required final action before resubmission

Correct all three exact strings and then run a whole-document proofread. Also fix the additional corrupted phrases identified in the grammar audit,

especially “excludes all children,” “electricity use of a of Japan,” and energy cost/and or.

Recommended response to the reviewer

We thank the reviewer. The four typographical issues have now been corrected throughout the manuscript: Data canters has been changed to Data centers; AI Workload Electrical Charactersitics to AI Workload Electrical Characteristics; Colling System Integration to Cooling System Integration; and the stray dollar sign before THD has been removed. We also completed a document-wide proofreading pass for merged words, duplicated fragments, missing spaces, and heading-format inconsistencies.

Comment 2.27

Reviewer comment

Currency appears as \$35–80/MWh, 35\$ to 80\$ per MWh, 200M\$ to 320M\$, and 20\$ per MWh. US/UK spellings are mixed within the same paragraphs. Section 1.3 'In congested regions' sentence has no main verb. Item 7 of Section 1.5 list reads as a leftover editing note. No nomenclature list despite heavy use of PUE, CFE, LCOE, LOLE, MILP, MPC, IBR, SCR, RoCoF, TRE, ADR, PFR. The abstract is also long and states no method.

Original manuscript

The original manuscript mixed currency conventions and U.S./U.K. spelling, contained a missing verb in the interconnection sentence, included an editing-note-like item, lacked a nomenclature list, and had an abstract that did not clearly report a review method.

Current revised manuscript

The current revision fixes some items: the Section 1.3 interconnection sentence now has a complete verb structure; item 7 of the framework list is a normal sentence; and the abstract now reports a PRISMA-informed review and 185-source evidence base.

However, the current manuscript still mixes \$ and USD, uses both U.S. and U.K. forms (modeling/modelling, decentralized/decentralised), has **no Nomenclature/Abbreviations section**, and the abstract is approximately **476 words**, substantially longer than the version described in the previous response letter.

Highlighted revision evidence

The abstract and several revised economic passages are highlighted, but the missing nomenclature and residual style inconsistencies are global issues rather than isolated highlighted changes.

Assessment

PARTIALLY ADDRESSED

Only part of the presentation pass is complete.

Required final action before resubmission

Standardize currency to USD value/unit; choose U.S. English or U.K. English and apply it globally; add a nomenclature/abbreviations section; shorten the abstract to the target journal limit while retaining method, n=185, critical findings, and framework scope. Do not claim the abstract is ~240 words in the response letter because the current abstract is ~476 words.

Recommended response to the reviewer

We agree. The manuscript has undergone a full presentation pass. Currency is now formatted consistently as USD <value>/<unit>, spelling has been standardized to the selected journal convention, the missing verb and editing-note artifact have been corrected, and a Nomenclature/Abbreviations section has been added for the principal technical acronyms. The abstract has also been shortened to comply with the journal's word limit while retaining the review method, the 185-source evidence base, the main critical findings, and the scope of the seven-component screening framework.

Comment 2.28

Reviewer comment

Missing: CRediT author contribution statement, declaration of competing interest, funding statement, data availability statement, declaration on the use of generative AI, and highlights. The reference list is not in a consistent Elsevier numbered style, and several entries lack volume, issue, or page numbers or give only a bare URL.

Original manuscript

The original manuscript did not contain the journal-required Highlights, CRediT statement, competing-interest declaration, funding statement, data-availability statement, generative-AI declaration, or a consistently formatted Elsevier-style reference list.

Current revised manuscript

Despite the claims in the older `Reviewer_Response_Letter.docx`, the **current uploaded manuscript does not contain** a Highlights section, Nomenclature section, CRediT author statement, Declaration of Competing Interest, Funding statement, Data Availability statement, or

Declaration on the Use of Generative AI. The reference list also still contains bare-URL/institutional entries and inconsistent metadata, and the current numbering appears to end at [178].

Highlighted revision evidence

There is no corresponding highlighted front/back-matter block in the current manuscript, confirming that these additions are absent from this upload.

Assessment

NOT ADDRESSED IN THE CURRENT UPLOAD

This comment is not addressed in the current file and must be completed before resubmission.

Required final action before resubmission

Add all required declarations using truthful information. Recommended templates:

Highlights — five concise bullets focused on the 185-source synthesis, seven-component screening framework, pack-vs-installed BESS distinction, unfirmed-vs-firm cost distinction, and site-specific reliability requirement.

CRedit — list only roles actually performed by each author.

Competing interests — use the journal's standard declaration if accurate.

Funding — state the actual funding/support; do not say “no external funding” if a grant supported the work.

Data availability — state that the review uses published sources; if a screening log/search export exists, state where it can be accessed.

Generative AI — disclose the actual tools and actual uses. Do not name a tool that was not used.

Then standardize every reference and reconcile all in-text citations.

Recommended response to the reviewer

We agree. The revised manuscript now includes the journal-required Highlights, CRediT author contribution statement, Declaration of Competing Interest, Funding statement, Data Availability statement, and Declaration on the Use of Generative AI. The reference list has also been standardized to the journal's numbered style, and all entries have been checked for complete bibliographic information and consistency with the in-text citations. The generative-AI declaration reports only the tools and purposes actually used by the authors, and the authors reviewed and take responsibility for the final manuscript content.

Comment 2.29

Reviewer comment

The title promises a critical review but the manuscript reads as a design guide, with almost every cited finding reported approvingly. Please either add appraisal of where the evidence is weak and where studies disagree, or adjust the title.

Original manuscript

The original paper was titled as a “critical review” but often read as a prescriptive design guide. It reported cited findings largely approvingly and presented the framework as though it generated universal design recommendations.

Current revised manuscript

The current manuscript is substantially more critical. Highlighted Section 2.6.2 explains why technical, economic, corporate, and vendor evidence has different boundaries and why no single evidence class establishes feasibility. Section 10 now emphasizes non-comparable cost boundaries. Section 13 explicitly states limitations. The abstract says the seven-component framework is a screening and decision-support framework rather than a site-specific design methodology, and Sections 5.3/7.3 disclaim universal optimization outputs.

Some prescriptive language remains, especially the Section 7 and Section 14 wording that calls the framework a framework “for designing, deploying, and operating” AI data centers. That wording should be softened to align completely with the revised critical-review positioning.

Highlighted revision evidence

Critical-appraisal and limitations text is heavily highlighted in Sections 2.6.2, 5.3, 7, 10, 13, and 14.

Assessment

SUBSTANTIALLY ADDRESSED

The reviewer’s concern has been substantially addressed, but the title-content alignment will be stronger if the remaining design-guide verbs are replaced by screening/synthesis language.

Required final action before resubmission

Change phrases such as `framework for designing, deploying, and operating` to `framework for screening and comparing energy-infrastructure pathways or structured decision-support framework`. Keep “design” only where describing the subject matter of the literature, not what this review itself claims to prescribe.

Recommended response to the reviewer

We agree that the title requires the manuscript to do more than summarize favorable findings. We have retained the title and strengthened the critical appraisal throughout the manuscript. The revised review now identifies where published studies use incompatible system boundaries, where evidence is geographically concentrated, where corporate metrics are self-reported, where cost benchmarks cannot be compared directly, where workload flexibility is constrained by commercial requirements, and where site-specific chronological analysis is necessary. The seven-component framework is now presented as a structured screening and decision-support framework rather than a prescriptive design methodology or universal technology recommendation.

Final synchronization checklist before sending the response letter

The following items should be completed in the **manuscript first**, then the response text above can be changed from conditional/recommended wording to definitive past tense:

1. Add verified search strings/search strategy details and the actual last search date for Comment 2.1.
2. Reconcile the 185-source evidence corpus with the current 178-entry bibliography and repair the dangling [186] citation.
3. Separate annual availability/unavailability from LOLE conceptually; do not tell the reviewer that 0.1 h/year equals 99.9886% — the reviewer's 99.9989% calculation is correct.
4. Make the AHP sensitivity procedure reproducible (recommended: $\pm 20\%$ relative one-at-a-time weight perturbation with

renormalization).

5. Add explicit `Source:` lines to **every** figure and table caption.
6. Fix the three still-present reviewer typos: `data canters`, `Charactersitics`, and `Colling`.
7. Standardize U.S./U.K. spelling and currency format, add a nomenclature list, and shorten the ~476-word abstract.
8. Add Highlights, CRedit, competing-interest, funding, data-availability, and truthful generative-AI declarations.
9. Remove reviewer-response narration from the manuscript, especially “The previous per-hour downtime value has been thus eliminated.”
10. Complete a final reference-to-claim audit after renumbering, especially all precise cost, reliability, corporate, and future-technology claims.

Bottom line

The new manuscript is **not merely a cosmetic revision**. Compared with the original, the highlighted changes substantially strengthen the review methodology, framework definition, storage-sizing discussion, economic boundaries, workload-flexibility treatment, critical appraisal, and limitations. The biggest remaining risk is that the **response letter currently overstates completion** of several reviewer requests that the actual manuscript still does not satisfy. Correcting the ten synchronization items above will make the response both technically defensible and consistent with the file the reviewer will actually read.